

A Matrix-Free Method for Large-Scale Quadratic Programs with Constraints

Thomas Schmelzer

Martin Stoll

Stephen Boyd

1 Overview

We consider quadratic programs of the form

$$\min_x \frac{1}{2}x^\top Px + c^\top x \quad \text{s.t.} \quad Ax \leq b, \quad Ex = d,$$

where $x \in \mathbb{R}^n$. We are interested in the regime in which n is large and the matrix $P \in \mathbb{R}^{n \times n}$ is not formed explicitly, but is available through matrix-vector products.

Typical examples include problems where

$$P = X^\top X + \lambda I,$$

with $X \in \mathbb{R}^{m \times n}$, or more generally where P is a linear operator that can be applied efficiently.

2 Matrix-Free Setting

We assume access to the following operations:

- $v \mapsto Pv$,
- $v \mapsto Av, v \mapsto A^\top v$,
- $v \mapsto Ev, v \mapsto E^\top v$.

No explicit formation of P , $X^\top X$, or any dense KKT matrix is required. The cost of applying P is assumed to be comparable to applying X and $X^\top$, for example:

$$Pv = X^\top(Xv) + \lambda v.$$

3 Optimality Conditions

The KKT conditions are

$$\begin{aligned} Px + c + A^\top \lambda + E^\top \nu &= 0, \\ Ax - b &\leq 0, \quad \lambda \geq 0, \quad \lambda_i(Ax - b)_i = 0, \\ Ex - d &= 0. \end{aligned}$$

Given an active set $\mathcal{A}$, we solve the equality-constrained subproblem:

$$A_{\mathcal{A}}x = b_{\mathcal{A}}, \quad Ex = d.$$

The corresponding KKT system is

$$\begin{bmatrix} P & A_{\mathcal{A}}^\top & E^\top \\ A_{\mathcal{A}} & 0 & 0 \\ E & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda_{\mathcal{A}} \\ \nu \end{bmatrix} = \begin{bmatrix} -c \\ b_{\mathcal{A}} \\ d \end{bmatrix}.$$

4 Krylov Subspace Solution

4.1 MINRES

The KKT matrix is symmetric but indefinite: the off-diagonal blocks $[A_{\mathcal{A}}^{\top}, E^{\top}]$ introduce negative eigenvalues, so the standard Conjugate Gradient (CG) method, which requires symmetric positive definiteness, cannot be applied directly to the full system. MINRES handles symmetric indefinite systems via short three-term recurrences and requires only matrix-vector products with the KKT operator:

$$(x, \lambda, \nu) \mapsto \begin{bmatrix} Px + A_{\mathcal{A}}^{\top} \lambda + E^{\top} \nu \\ A_{\mathcal{A}} x \\ Ex \end{bmatrix}.$$

No matrix factorisation is required. Warm starts are used between active-set iterations.

4.2 Conjugate Gradient in the Null Space

An alternative that recovers a symmetric positive definite (SPD) system is to eliminate the equality constraints explicitly and apply CG to the resulting reduced problem. We describe the procedure in detail.

Setup. Collect all currently active equality constraints (active-set inequalities and hard equalities) into the single system

$$Bx = f, \quad B = \begin{bmatrix} A_{\mathcal{A}} \\ E \end{bmatrix} \in \mathbb{R}^{s \times n}, \quad f = \begin{bmatrix} b_{\mathcal{A}} \\ d \end{bmatrix} \in \mathbb{R}^s, \quad (1)$$

where $s = |\mathcal{A}| + \text{rows}(E)$ is the total number of active constraints. We assume B has full row rank ($s < n$) and $P \succ 0$.

Null-space basis via QR. Compute the full QR factorisation of $B^{\top} \in \mathbb{R}^{n \times s}$:

$$B^{\top} = Q \begin{bmatrix} R \\ 0 \end{bmatrix}, \quad Q = [Q_1 \mid Q_2], \quad (2)$$

where $Q \in \mathbb{R}^{n \times n}$ is orthogonal, $R \in \mathbb{R}^{s \times s}$ is upper triangular, $Q_1 \in \mathbb{R}^{n \times s}$ spans the column space of $B^{\top}$, and $Q_2 \in \mathbb{R}^{n \times (n-s)}$ spans the orthogonal complement. Because $BQ_2 = (Q_2^{\top} B^{\top})^{\top} = 0$, the columns of Q_2 form an orthonormal basis for $\ker(B)$.

Particular solution. A particular solution x_p satisfying $Bx_p = f$ is obtained from the QR factors by triangular back-substitution:

$$x_p = Q_1 R^{-\top} f.$$

Constraint elimination. Every vector satisfying $Bx = f$ can be decomposed as

$$x = x_p + Q_2 \tilde{x}, \quad \tilde{x} \in \mathbb{R}^{n-s}, \quad (3)$$

since $B(x_p + Q_2 \tilde{x}) = f + BQ_2 \tilde{x} = f$. Substituting (3) into the quadratic objective and differentiating with respect to the free variable $\tilde{x}$:

$$\frac{\partial}{\partial \tilde{x}} \left[\frac{1}{2} (x_p + Q_2 \tilde{x})^{\top} P (x_p + Q_2 \tilde{x}) + c^{\top} (x_p + Q_2 \tilde{x}) \right] = 0$$

yields the $(n - s) \times (n - s)$ reduced linear system

$$\underbrace{Q_2^\top P Q_2}_{\hat{P}} \tilde{x} = -Q_2^\top (P x_p + c). \quad (4)$$

The reduced matrix $\hat{P} = Q_2^\top P Q_2$ is SPD: $P \succ 0$ implies $\tilde{x}^\top \hat{P} \tilde{x} = (Q_2 \tilde{x})^\top P (Q_2 \tilde{x}) > 0$ for all $\tilde{x} \neq 0$ because Q_2 has full column rank. CG is therefore directly applicable to (4).

Matrix-free reduced operator. Each application of $\hat{P}$ to a vector $\tilde{v} \in \mathbb{R}^{n-s}$ proceeds without forming $\hat{P}$ explicitly:

1. $u \leftarrow Q_2 \tilde{v}$ (lift to $\mathbb{R}^n$; cost $O(n(n - s))$)
2. $w \leftarrow P u = X^\top (X u) + \lambda u$ (apply P matrix-free; cost $O(Tn)$)
3. $\hat{w} \leftarrow Q_2^\top w$ (project back; cost $O(n(n - s))$)

The dominant per-iteration cost is therefore $O(Tn + n^2)$.

QR setup cost. Computing the full orthogonal factor Q by applying s Householder reflectors to the $n \times n$ identity costs $O(n^2 s)$ per active-set step. When many inequality constraints are active—e.g. long-only with $s = O(n)$ —this reaches $O(n^3)$ and dominates the per-iteration cost, making constraint-eliminated CG slower than MINRES in that regime. MINRES, by contrast, incurs no such setup cost and has per-iteration cost $O(Tn + sn) = O(Tn)$ for $s \ll T$.

Special case: single budget constraint. When $B = \mathbf{1}^\top$ ($s = 1$), the null-space basis Q_2 is the Householder complement of $\hat{e}_1 = \mathbf{1}/\|\mathbf{1}\|$. The maps $Q_2 \tilde{v}$ and $Q_2^\top w$ can be applied implicitly via the rank-one reflector $I - 2\hat{e}_1 \hat{e}_1^\top$ in $O(n)$, reducing the per-iteration cost from $O(n^2)$ to $O(Tn)$ and the QR setup to $O(n)$. This is the approach used in the minimum-variance companion paper for the single budget constraint.

Recovering the primal solution. After CG converges to $\tilde{x}^*$, the full weight vector is recovered via (3):

$$x^* = x_p + Q_2 \tilde{x}^*.$$

5 Active-Set Method

We use a standard active-set method:

1. Initialize with a feasible point and active set $\mathcal{A}$.
2. Solve the equality-constrained subproblem.
3. If a constraint is violated, add it to $\mathcal{A}$.
4. If a multiplier is negative, remove the corresponding constraint.
5. Repeat until optimality conditions are satisfied.

6 Preconditioning

Performance depends critically on preconditioning. Possible choices include:

- Diagonal or block-diagonal approximations,
- Ridge regularization $P + \gamma I$,
- Low-rank approximations $P \approx BFB^\top + D$.

The preconditioner must be inexpensive and matrix-free.

7 Computational Cost

Let T_P denote the cost of computing Pv , and T_A the cost of applying A .

Each Krylov iteration costs

$$O(T_P + T_A).$$

The total cost is

$$O(N_{\text{outer}} \cdot N_{\text{Krylov}} \cdot (T_P + T_A)).$$

For $P = X^\top X$, we have $T_P = O(\text{nnz}(X))$. No cubic operations are required.

8 Numerical Considerations

- Conditioning of P and the KKT system affects convergence.
- Ridge regularization improves stability.
- Scaling of constraints is important.
- Moderate accuracy is usually sufficient.

9 Comparison with Existing Solvers

Clarabel. Clarabel is a modern interior-point method designed for conic optimization. It solves a sequence of KKT systems using sparse direct factorizations. It is robust and highly accurate, but memory and runtime scale with explicit matrix formation and factorization. In large-scale matrix-free settings, this becomes limiting.

OSQP. OSQP is an ADMM-based operator splitting solver. It factorizes a fixed KKT matrix once and reuses it across iterations, making it efficient for medium-scale problems and repeated solves. However, convergence is linear and performance degrades for ill-conditioned problems or high accuracy requirements. It also requires explicit matrix representations.

10 Discussion

The method avoids forming dense matrices and relies only on matrix-vector products. It scales well to large problems but depends on good preconditioning.

Compared to factor models, it does not require low-rank structure, but may be less efficient when such structure is strong and exploitable.

11 Conclusion

A matrix-free active-set method combined with Krylov solvers provides a practical approach for large-scale constrained quadratic programs.